Sea Winds Predictions, Phase 2 Report

Michael Ibrahim

Executive summary

The competition joins three decisions: forecasting wind speed and direction on the 1.3 km grid, selecting a legal site and layout for 55 IEA 22 MW turbines, and converting the wind forecast into farm-energy quantiles. Reproducibility code is available at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code`. The repository provides two Python entry points: `train.py` creates one artifact bundle from the permitted inputs, and `inference.py` loads that bundle to generate `predictions.csv` and `submission.json`. No previous submission is an input.

The forecast is a compact statistical correction of the supplied atmospheric predictors. Speed is estimated with lead-specific quantile models; direction is corrected in vector space to respect circular geometry; two component models map coarse wind to the target grid. A small set of context and external-trajectory corrections is applied only where historical replay supports it. The competition scorer matched all 4,196,640 submitted rows. The final interval scores were 9.511, 17.124, and 16.835 for speed and 76.400, 335.571, and 315.183 for direction at one, seven, and fourteen days, respectively. The rank-based primary score was 1.497276.

The farm centre is 54.109676°N, 0.952316°E. Its supplied-grid depth is 44.14 m. All 55 turbines satisfy the zone, depth, footprint, and spacing rules. Five-year replay with the prescribed PyWake configuration gives a mean capacity factor of 50.36%, mean annual energy production of 5,337.7 GWh, and mean wake loss of 7.10%. At the same centre, the selected layout improves mean capacity factor by 0.49 percentage points and reduces mean wake loss by 0.89 percentage points relative to a valid rectangular layout. The supplied cost model gives capital expenditure of EUR 4,478.4 million and mean-weather levelized cost of energy of EUR 88.69/MWh.

Evidence is reported according to how it was obtained. Historical validation is used to select forecast components. Competition-platform scores measure the final forecast against hidden target values. Rule compliance is checked exactly from the generated files. Cost, revenue, and bidding quantities are scenario calculations based on stated assumptions. Keeping these four forms of evidence separate prevents a validation result or a modeled financial quantity from being presented as an observed final outcome.

1 Task, data, and evidence boundary

1.1 Required outputs

The forecast output contains lower, median, and upper estimates for wind speed and direction at one, seven, and fourteen days. The scored footprint has 43,715 points. With eight inference windows, three lead times, and four forecast hours, the required row count is

$$43{,}715 \times 8 \times 3 \times 4 = 4{,}196{,}640.$$

The siting output specifies one centre and 55 turbine coordinates. Each turbine must remain inside the 15 km by 15 km layout box, every pair must be at least 1,420 m apart, and the centre depth must not exceed 50 m. The turbine type and total capacity are fixed by the rules [1, 2].

1.2 Declared data use

Training uses the official 2016–2020 high-resolution target, low-resolution reanalysis, supplied European Centre for Medium-Range Weather Forecasts High Resolution Forecast tables, bathymetry, and static grids [3]. The definitive eight-window release is used only as an unlabeled inference input [4]. One external source is used for forecasting: causal GraphCast trajectories from the WeatherBench 2 archive. Here, causal means that each trajectory was initialized using information available at its issue time and contains no later observations. GraphCast is an ERA5-trained model and is sampled only from runs initialized at the relevant issue time [9, 10].

Historical target values are never available to `inference.py`. Reported evidence is separated into four categories:

- **Historical validation**: comparisons on reserved official target rows from 2016–2020.
- **Definitive forecast evaluation**: the six scores returned by the competition platform.
- **Deterministic audits**: row coverage, quantile ordering, siting legality, and energy bounds.
- **Scenario analysis**: economics and bidding calculations whose assumptions are stated explicitly.

1.3 Forecast score

Throughout the report, $\widehat{Q}_p$ denotes a conditional quantile at probability p. The speed interval is $[\widehat{Q}_{0.05}, \widehat{Q}_{0.95}]$, with median $\widehat{Q}_{0.50}$. For observation y, endpoints l and u, and nominal error probability $\alpha = 0.10$, the linear interval score is

$$S_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}.$$

This proper score rewards narrow intervals only when they retain coverage [5, 6]. Direction uses the competition's circular counterpart because 0° and 360° represent the same direction. Lower scores are better. The primary score is rank based and is not the mean of the six raw scores.

2 Reproducible code and end-to-end execution

2.1 Code availability

The public repository at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code` contains `train.py`, `inference.py`, the pinned Python requirements, data-placement instructions, third-party licensing disclosures, this methodology report, and the final output archive. Competition datasets and external forecast fields are not redistributed. The README specifies where those inputs must be placed and gives the complete commands needed to regenerate the artifact and outputs.

The separation between training and inference is deliberate. Training may read labeled official rows from 2016–2020 and writes fitted parameters, validation decisions, the selected farm geometry, and the power-conversion policy. Inference reads only that artifact and unlabeled forecast inputs. It does not read historical targets, validation tables, or a previous `predictions.csv`. This makes the causal boundary inspectable in code rather than relying on a narrative claim.

2.2 Execution stages

Figure 1 gives the full computational path. The upper branch constructs forecast features, fits the probabilistic models, and retains optional corrections only after chronological, coverage, worst-year, and diagnostic-group checks. The middle branch screens legal centres before exact farm and layout replay. Both branches write one artifact. Inference then reads that artifact and the unlabeled official windows, processes one window at a time, checks physical and structural constraints, and writes the two required outputs.

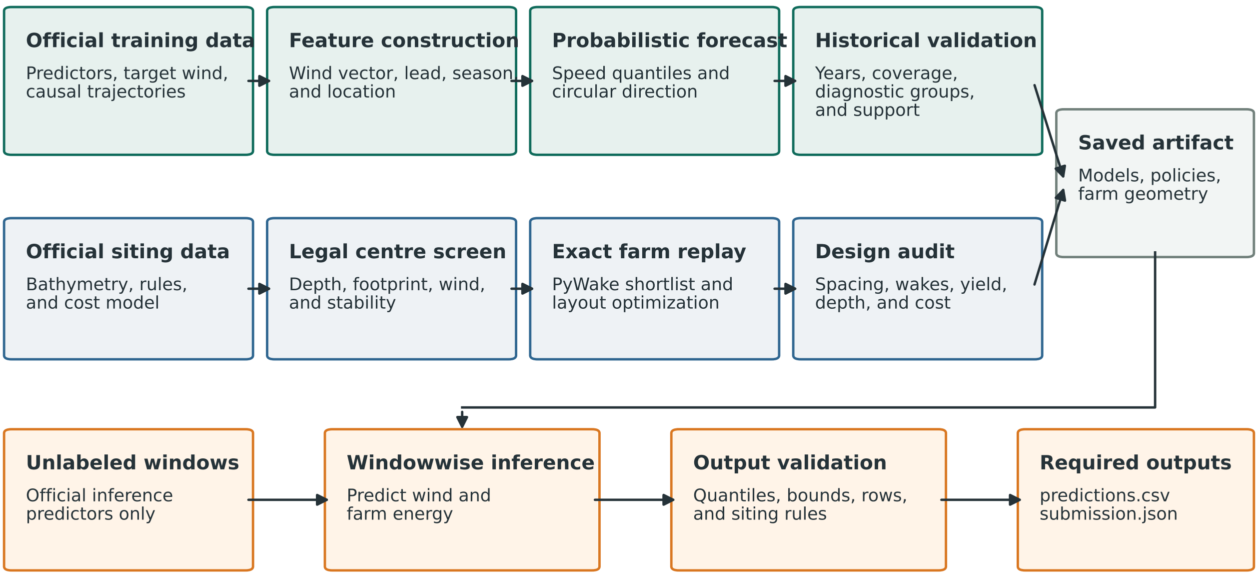

Figure 1: End-to-end generation pipeline. Labeled wind targets are confined to training and validation. Inference reads only the saved artifact and unlabeled official predictors.

The production artifact contains 29 fitted estimators: nine lead-by-quantile speed models, six support-gated quantile refinements, five context endpoint models, four circular direction residual models, one shared spatial direction model, two conditional direction endpoint models, and two component downscalers. Every estimator is single-threaded. The number is reported to define the actual computational object, not as a performance claim.

The repository README records the exact commands and final parameter values. Reproduction requires one call to `train.py`, followed by one call to `inference.py`; neither command invokes another project program. The final configuration uses a six-day training sample, a 0.90 speed-coverage target, one-day speed-width scale 0.75, maximum seven-day centre-policy weight 0.4, one-day context-blend scale 1.25, and a 145°cap on the fourteen-day direction half-width. These values are serialized or passed explicitly, so the generated result does not depend on an undocumented interactive step.

3 Forecasting method

3.1 Speed and spatial downscaling

The core speed model is quantile model-output statistics implemented with LightGBM [7]. Separate models estimate $\widehat{Q}_{0.05}$, $\widehat{Q}_{0.50}$, and $\widehat{Q}_{0.95}$ at each lead time. Direct quantile fitting aligns training with the interval-score objective and avoids assuming symmetric or constant residual variance.

Predictors are limited to quantities with a physical or operational interpretation: supplied eastward and northward wind components, their magnitude and simple interactions, issue hour, seasonal sine–cosine terms, latitude, longitude, elevation, and coast distance. Wind components retain flow orientation; seasonal harmonics avoid an artificial discontinuity between the end and beginning of the year. Two additional regressors map coarse eastward and northward wind to the 1.3 km target grid. A residual calibration estimated from reserved historical validation data adjusts interval width. Finally, endpoints are sorted and clipped so that $0 \le \widehat{Q}_{0.05} \le \widehat{Q}_{0.50} \le \widehat{Q}_{0.95}$.

3.2 Circular direction

Angular residuals cannot be fitted safely as ordinary numbers because a small error across north appears numerically close to 360°. The model therefore converts direction to sine and cosine components, learns residual corrections in that space, blends vectors, and converts the result back to degrees. This is standard directional-statistics practice [8]. Direction interval centres and widths are lead specific.

3.3 Different information at each lead

The available signal changes with forecast lead, so the three branches are not forced to share one correction.

- **One day**: supplied forecast fields provide the main signal. Lagged context corrects speed endpoints. Causal GraphCast wind at +24 to +42 hours supplies an additional direction trajectory.
- **Seven days**: supplied seven-day fields remain the anchor. A small context endpoint model and a gated external trajectory correction are active only on historically supported rows.
- **Fourteen days**: the supplied forecast does not extend to fourteen days. The branch combines the latest available forecast state with seasonal information and historically calibrated endpoint transformations. This is the least direct branch and is treated most conservatively.

The original contribution is not a new atmospheric simulator. It is the integration of lead-specific probabilistic calibration, circular correction, high-resolution vector mapping, and causal trajectory information under one reproducible selection and inference path. Its practical purpose is to improve the interval endpoints without building a large ensemble.

4 Validation and forecast results

4.1 Validation protocol

Validation asks one practical question: does an added forecast component improve predictions on historical observations reserved for evaluation? The same four-step procedure is used for each speed and direction target:

1. Reserve one or more historical periods as validation data for the component being tested.
2. Produce two forecasts for exactly the same validation rows: one from the current pipeline without the new component and one from the candidate pipeline with it.
3. Score both forecasts with the competition interval score. A lower score indicates a better forecast. For speed, interval coverage must also remain at or above the stated 90% target.
4. Retain the component only if the total score improves and no available validation year or populated diagnostic group becomes worse. The diagnostic groups divide the rows by location, wind speed, direction, interval width, and signed error so that an average gain cannot conceal a clear localized loss.

The final inference labels are never read during this process. Their unlabeled predictors are used only to check whether a proposed correction would activate within conditions represented in the historical data. Table 1 reports one paired comparison for every scored target. Validation periods differ because the six components use different leads and activation conditions. The rows therefore document six individual selection decisions and should not be added together as one overall validation score.

Target	Added component	Validation period	Without	With	Δ score
Speed, 1 day	Context endpoints	2020	13.040	11.468	−1.572
Speed, 7 days	Cell endpoint policy	2016–2020	18.143	17.157	−0.987
Speed, 14 days	Cell endpoint policy	2019–2020	21.943	17.013	−4.931
Direction, 1 day	Causal trajectory blend	2018, 2020	n.a.	n.a.	−4.857
Direction, 7 days	Conditional endpoints	2017–2020	n.a.	n.a.	−2.331
Direction, 14 days	February 25 width	2019, 2020	n.a.	n.a.	−208.123 to −168.594

Table 1: Controlled historical comparisons retained in the final artifact. "Without" and "With" refer to the added component in the second column; n.a. means that the absolute intermediate scores were not retained. Negative score change is better. The paired score changes were retained for every component. The fourteen-day direction row applies only to the four February 25 forecast hours selected by the validation rule.

Coverage was retained for the two multi-year speed comparisons. Seven-day coverage changed from 94.20% to 93.95%; fourteen-day coverage changed from 94.58% to 91.75%. Both remain above the nominal 90% level, so the score reductions are not explained by intervals that are too narrow. The one-day direction changes were -2.329 in 2018 and -7.385 in 2020. The seven-day direction correction improved every validation year, from -2.007 to -2.785, with no populated diagnostic group worsening beyond numerical tolerance. The February-only fourteen-day policy improved every validation year and forecast-hour combination. These tests are intentionally conservative, although a correction acting on a small subset of rows can have a larger local effect than its effect on the complete target.

Two examples clarify the selection rule. A stronger one-day external speed blend improved its mean score but failed localized issue-time checks and was rejected. A separate fine-grid speed-residual model failed its transfer gate and is not present in the production artifact. These rejected branches matter because they show that aggregate development score alone did not determine the final pipeline.

4.2 Interpretation of the fitted signal

Normalized LightGBM split gain is used as a descriptive audit of the fitted speed models, not as a causal explanation. At one day, the supplied forecast-speed variables account for 50.6% of normalized gain; location, vector components, and seasonal terms contribute 14.3%, 11.5%, and 11.3%. At fourteen days, direct forecast speed contributes only 5.4%, while seasonal terms, location, and wind-vector components together contribute 76.5%. This shift is consistent with the information boundary: direct forecast guidance is strongest at short lead, whereas the long-lead model relies more heavily on climatological and spatial structure.

Activation fractions provide a second, model-independent interpretation. The accepted one-day context correction modifies 61.87% and 74.97% of rows in the two retained validation folds; the seven-day correction is deliberately sparse at 6.25%; and the fourteen-day endpoint policy acts on 84.38% of its evaluated rows. These fractions explain why a small seven-day aggregate gain can still be credible and why the fourteen-day policy required stronger year and diagnostic-group checks.

4.3 Definitive evaluation

The competition scorer matched 4,196,640 of 4,196,640 rows. Table 2 reports the complete definitive result.

Interval score	One day	Seven days	Fourteen days
Wind speed	9.511	17.124	16.835
Wind direction	76.400	335.571	315.183
Platform primary score			1.497276

Table 2: Definitive forecast scores returned by the competition platform. Lower values are better; the primary score is rank based.

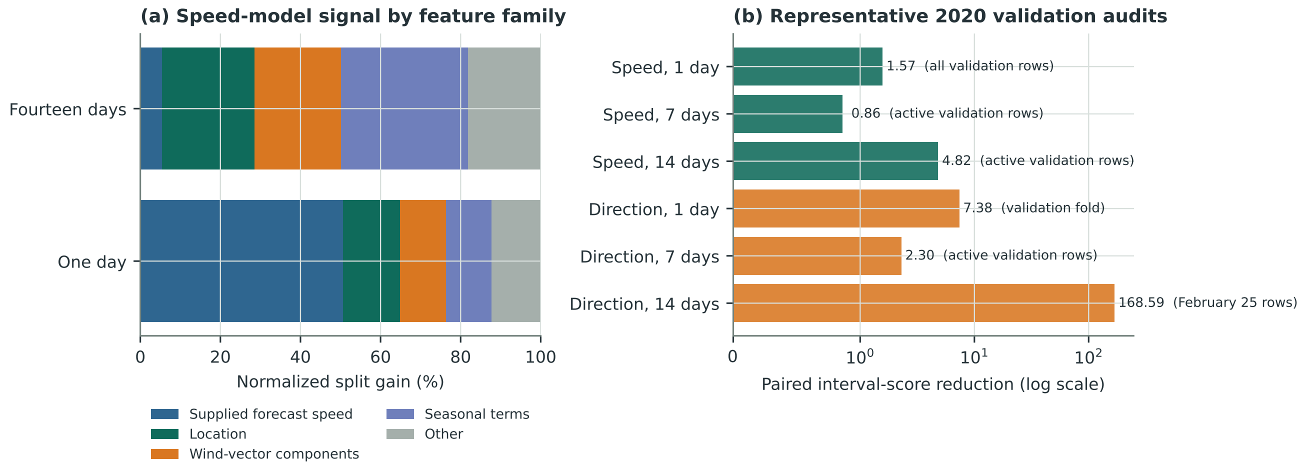

Figure 2: Model interpretation and validation audit. Panel (a) groups normalized split gain by feature family; it describes model use rather than causality. Panel (b) shows a representative paired reduction on 2020 validation rows for each target branch and its component-specific activation mask. The logarithmic axis makes the sparse February-only fourteen-day direction audit visible without hiding the other five; magnitudes are not compared across masks.

The result is uneven. One-day direction is the strongest directional component, whereas seven- and fourteen-day direction remain the largest forecast weakness. The report does not infer calibration or generalization from the platform score alone.

5 Wind-farm siting and power conversion

5.1 Centre selection

The centre search first evaluates gross wind on the supplied grid because wake simulation at every cell would be unnecessarily expensive. Of 37,835 weather cells screened, 23,406 pass the geographic and depth rules. Fourteen candidates representing high mean wind, stable worst-year wind, balanced performance, and depth margin receive exact five-year PyWake replay. The final run retains the selected reference centre because no challenger improves mean and worst-year capacity factor while also satisfying the neighborhood, wake, depth, and economic gates. This is a robust selection rule, not proof of a global optimum.

5.2 Layout selection

At the selected centre, 16 valid layouts are replayed with the prescribed IEA 22 MW turbine and Bastankhah-Gaussian wake model [11]. The final geometry comes from constrained coordinate refinement and uses the available square while preserving spacing. Figure 4 shows the 55 submitted positions.

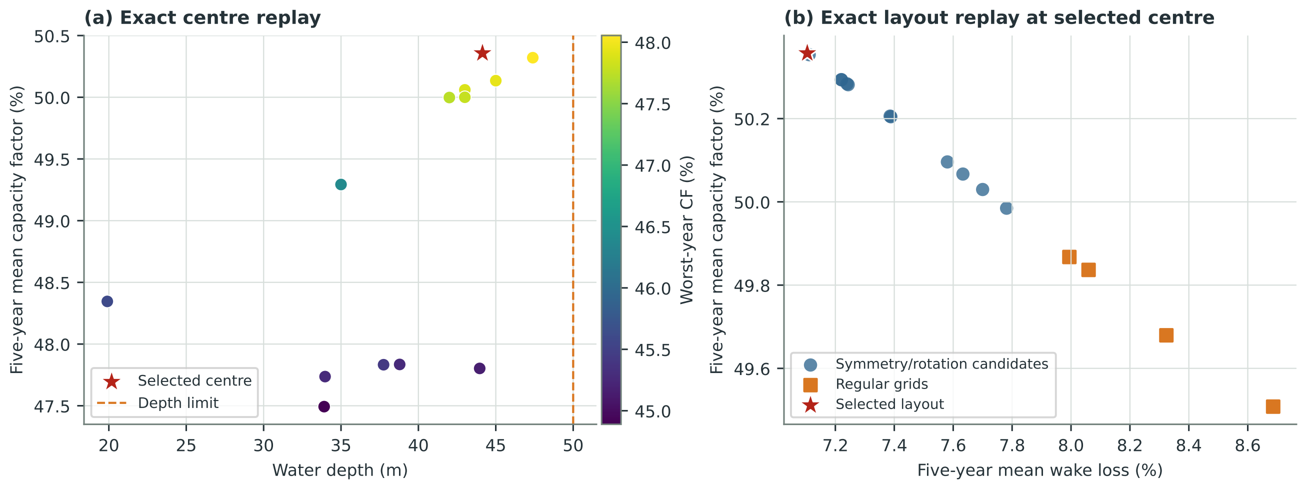

Figure 3: Exact five-year siting evidence. Panel (a) shows the 14 shortlisted legal centres; colour records worst-year capacity factor, so the selected point combines high mean yield with cross-year resilience while retaining depth margin. Panel (b) shows all 16 valid layout candidates at the selected centre; moving upward and left means more yield and less wake loss.

The legal audit is deterministic and is summarized in Table 3. The centre is inside the allowed zone and at 44.14 m depth. The layout uses almost all of the permitted north–south span, so coordinates are retained at full numer-

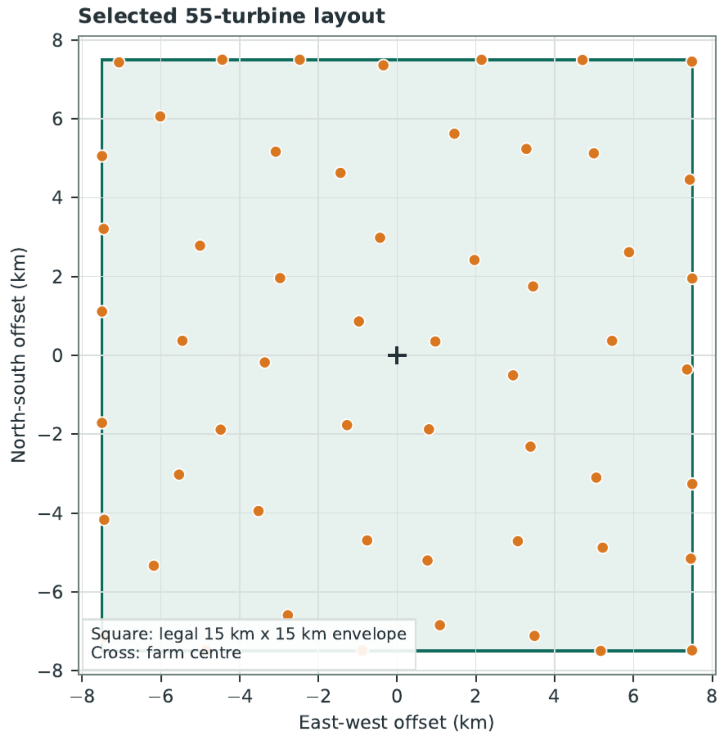

Figure 4: Submitted turbine coordinates relative to the farm centre. The dashed square is the permitted layout envelope.

ical precision in `submission.json`; rounding the outer coordinates would unnecessarily consume the remaining 0.06 m margin.

Constraint	Observed	Requirement	Margin
Turbine count	55	exactly 55	0
Installed capacity	1,210 MW	exactly 1,210 MW	0
Centre depth	44.14 m	at most 50 m	5.86 m
Maximum $\lvert x \rvert$	7,495.61 m	at most 7,500 m	4.39 m
Maximum $\lvert y \rvert$	7,499.94 m	at most 7,500 m	0.06 m
Minimum pairwise spacing	1,554.64 m	at least 1,420 m	134.64 m

Table 3: Deterministic legality audit of the submitted centre and coordinates.

The selected centre is 69.07 km from the nearest coast in the supplied distance layer. Distance and depth enter the cost model through export-cable and foundation premiums. Economics is therefore used as a secondary discriminator between near-equal capacity-factor candidates; it does not displace a materially better yield candidate.

Coordinate refinement evaluated 340 legal moves and accepted 26. Against the pre-refinement geometry, the final coordinates improved mean capacity factor by 0.013 percentage points, worst-year capacity factor by 0.019 percentage points, and every individual training year. In a 2,000-replicate monthly block bootstrap, every replicate retained a positive capacity-factor change; the first-percentile change was 0.009 percentage points. The layout also passed the mean-yield, worst-year-yield, and wake gates at all 31 nearby legal centres tested. These small but consistently signed changes support robustness; they are not presented as a material project-level yield increase.

Table 4 isolates the value of the layout by holding the centre and five-year weather replay fixed. Relative to a valid rectangular 7-by-8 arrangement with one unused position, the submitted geometry adds 51.9 GWh/year on average and reduces wake loss by 0.89 percentage points.

Layout at selected centre	Mean CF	Worst CF	Mean AEP (GWh/year)	Mean wake
Valid rectangular 7-by-8	49.87%	47.73%	5,285.8	7.99%
Submitted geometry	50.36%	48.18%	5,337.7	7.10%
Change	+0.49 pp	+0.45 pp	+51.9	−0.89 pp

Table 4: Same-centre, same-weather comparison that isolates the submitted layout. CF is capacity factor; pp denotes percentage points.

The annual spread is material: capacity factor ranges from 48.18% to 54.95%, while wake loss remains between

Replay year	AEP (GWh)	Capacity factor	Wake loss
2016	5,106.9	48.18%	7.31%
2017	5,824.1	54.95%	6.93%
2018	5,279.1	49.80%	7.00%
2019	5,327.4	50.26%	7.31%
2020	5,150.7	48.59%	6.97%
Mean	5,337.7	50.36%	7.10%

Table 5: Exact annual PyWake replay of the selected site and layout using the supplied historical wind.

6.93% and 7.31%. A directional stress test shifts the complete wind rose from $-45°$ to $+45°$; capacity factor remains between 50.23% and 50.38%, while maximum wake loss is 7.32%. This test shows that the irregular geometry is not narrowly tuned to one wind-rose orientation.

A second benchmark evaluates the combined centre-and-layout decision on public training data. A plain 7D grid at the public reference centre, snapped to 53.4982°N and 1.4928°E, gives 47.08% mean capacity factor, 4,989.9 GWh mean AEP, and 9.70% mean wake loss under the same five-year replay. The selected design gives 50.36%, 5,337.7 GWh, and 7.10%, respectively. This is explicitly a public-data proxy, not the organizer's hidden real-wind baseline; it is included because its inputs and simulation path are reproducible.

5.3 Farm-energy quantiles

For each inference time, `inference.py` extracts wind quantiles at the selected centre, applies the prescribed power-law shear from 125 m to the 170 m hub height, and runs the submitted geometry through the turbine power curve and wake model. The three paths are sorted after simulation and clipped to the physical six-hour range $[0, 7{,}260]$ MWh. The final JSON contains 96 values for each quantile. Their means are 738.9, 3,001.8, and 6,735.1 MWh for the lower, median, and upper paths.

This construction guarantees consistency with the submitted wind forecast and farm geometry. It does not guarantee calibrated farm-power coverage because a common median direction is used for all three speed paths and the joint speed–direction distribution is not modeled.

6 Economics and forecast value

6.1 Construction and operating cost

The supplied Phase 2 cost model is evaluated at the selected centre. Table 6 shows the complete decomposition used in the report. The intensity column is capital cost divided by the farm's full installed capacity of 1,210,000 kW. For example, 1,030 EUR per installed kW corresponds to EUR 1,246.3 million for the 1.21 GW farm. Depth and export distance are the site-dependent terms; no cost item is replaced with an external estimate.

Capital component	EUR per installed kW	EUR million
Turbines and towers	1,030.00	1,246.3
Foundations, including depth premium	812.10	982.6
Array electrical system	350.00	423.5
Export cable, including distance premium	552.56	668.6
Installation	350.00	423.5
Development	310.00	375.1
Contingency	296.47	358.7
Total capital expenditure	**3,701.13**	**4,478.4**

Table 6: Construction-cost result from the supplied Phase 2 cost model. Costs in the middle column are normalized by 1.21 GW of installed nameplate capacity.

Annual operating expenditure is EUR 123.1 million. Using mean historical energy, the model gives a levelized cost of energy of EUR 88.69/MWh, comprising 65.63 EUR/MWh of capital recovery and 23.06 EUR/MWh of operation. Replacing mean energy with the lowest historical year raises the result to 92.70 EUR/MWh. This four-euro range is the direct financial consequence of the recorded interannual wind variability under the supplied assumptions.

Two independent references provide a plausibility check, not a replacement cost model. The 2025 *Guide to an Offshore Wind Farm* uses a representative 1 GW UK project at 50 m depth and 75 km distance and reports capital cost of about GBP 3.47 million/MW and annual operating cost of GBP 85,000/MW [15]. The selected site's

EUR 3.70 million/MW capital cost and EUR 101,721/MW-year operating cost are of comparable order. A direct numerical match is neither expected nor claimed because currencies, price years, turbine ratings, grid boundaries, and contracting scope differ. NREL's 2025 United States fixed-bottom assessment places most sites above USD 6,000/kW and operating cost above USD 83/kW-year; it also finds that distance from coast, port, and interconnection increases both costs [17, 16]. The higher United States capital estimate shows why geographic and accounting boundaries matter. Taken together, the literature supports the model's dependence on depth and distance but validates only order of magnitude, not project bankability.

6.2 Historical market context and long-term viability

The five-year mean AEP is 5.338 TWh. For scale, public Energy-Charts data report 100.39 TWh of Dutch load in 2022, so the modeled farm output equals 5.32% of that annual quantity [20]. This is a scale comparison, not a grid-connection or market-share claim. The same source gives a 2022 Dutch day-ahead mean of 241.92 EUR/MWh and median of 217.00 EUR/MWh. That unusually high and volatile year is useful for a historical stress context, but it is not adopted as a 25-year price forecast.

For the 32 available one-day definitive blocks, the median farm-energy forecast totals 121.193 GWh. Aligning each six-hour block with the public 2022 hourly day-ahead price gives a gross committed value of EUR 23.751 million and an energy-weighted capture price of 195.98 EUR/MWh, compared with an unweighted 229.93 EUR/MWh over those hours. These figures value the forecast at historical prices; without realized farm output and settlement prices, they are not realized revenue.

Long-term profitability is therefore presented as a scenario. At mean annual energy of 5,337.7 GWh, constant energy prices of 70, 90, and 110 EUR/MWh imply gross annual revenue of EUR 373.6, 480.4, and 587.1 million. With capital expenditure C_0, annual energy E, annual operating cost O, energy price P, discount rate $r = 0.06$, and a 25-year operating life, the calculation is

$$\mathrm{NPV}(P) = -C_0 + \sum_{t=1}^{25} \frac{PE - O}{(1+r)^t}.$$

The resulting pre-tax NPVs are EUR $-1{,}275.5$, 89.2, and 1,453.9 million. Figure 5 adds one-at-a-time sensitivity to discount rate, capital cost, historical annual energy, project life, and operating cost. Energy price has the largest tested range, followed by discount rate. The project is therefore close to break-even at 90 EUR/MWh but not robustly profitable under the tested assumptions. NPV excludes tax, degradation, curtailment, construction delay, financing structure, and residual value.

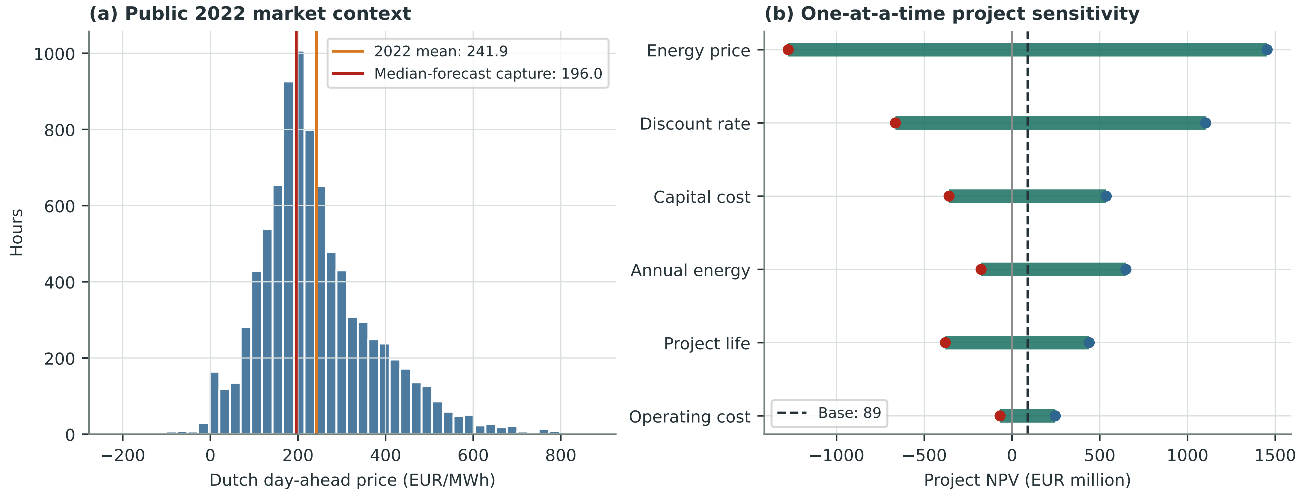

Figure 5: Historical market context and project sensitivity. Panel (a) shows all hourly Dutch day-ahead prices in 2022 and the capture price obtained by weighting the 32 available definitive one-day blocks by median forecast energy. Panel (b) varies one assumption at a time around the 90 EUR/MWh, 6%, 25-year case; endpoints are scenario bounds, not confidence intervals.

The supplied operating-cost value is an aggregate allowance, not a component-level maintenance model. Offshore operations combine scheduled work, unscheduled repair, marine access, insurance, and asset management; the industry guide reports that access restrictions alone can lower availability from roughly 98% to 95–98% [18]. NREL identifies condition-based maintenance, remote inspection, automated repair, personnel transfer, and better weather forecasting as credible cost-reduction mechanisms, while also emphasizing the lack of transparent public failure and repair data needed to quantify each mechanism [19]. The permitted inputs contain no turbine condition stream, failure history, vessel availability, or repair duration. A numerical maintenance saving would therefore be invented. The defensible strategy is operational: monitor vibration, temperature, oil, power-curve residuals, and alarms; update component failure hazards; and schedule low-wind vessel windows subject to access forecasts. No cost saving from this unobserved extension is credited to the NPV.

6.3 Quantile bidding

If shortfall and excess penalties are π^+ and π^-, respectively, the cost-minimizing bid is the predictive quantile [12]

$$\tau^* = \frac{\pi^-}{\pi^+ + \pi^-}.$$

For the supplied illustrative penalties of 90 EUR/MWh for shortfall and 30 EUR/MWh for excess, $\tau^* = 0.25$. The 25th-percentile energy bid is obtained by linear interpolation between the literal `predicted_q05` and `predicted_q50` arrays. Across the 32 one-day six-hour blocks, its mean is 2,487.1 MWh and the total commitment is 79.59 GWh.

To quantify the information value without pretending that outcomes are observed, the submitted 5th, 50th, and 95th percentiles define a piecewise-linear predictive quantile function; the two five-percent tails are linear continuations clipped to the physical range. Integrating the asymmetric loss over that distribution gives an expected imbalance cost of EUR 2.504 million for a median bid and EUR 1.880 million for the 25th-percentile bid. The quantile policy therefore reduces ex-ante expected imbalance cost by EUR 0.624 million, or 24.9%, across the available blocks under the stated penalties. This is a reproducible model-based value calculation, not a realized profit backtest. Realized farm production and imbalance settlement prices for the definitive blocks are unavailable.

7 Frugality, transparency, and verification scope

7.1 Compute footprint

Training and siting replay took 4.840 CPU-hours; inference and archive generation took 0.272 CPU-hours. No GPU was used. The measured machine used an Intel Core i7-7700HQ, and all estimators and numerical libraries were restricted to one thread. Table 7 converts wall-clock time using the processor's nominal 45 W thermal design power [13]. This is a CPU-package proxy, not whole-system electricity metering. Carbon uses a declared sensitivity of 0.45 $kgCO_2e$/kWh [14].

Stage	CPU-hours	GPU-hours	kWh	$kgCO_2e$
Training, validation, and siting	4.840	0	0.218	0.098
Inference and packaging	0.272	0	0.012	0.006
Total	**5.112**	**0**	**0.230**	**0.104**

Table 7: Measured runtime and bounded CPU-package energy and carbon estimates.

The estimate excludes memory, storage, display, networking, cooling, and the upstream cost of producing the public GraphCast archive. Inference processes one window at a time and successful training deletes temporary checkpoints.

7.2 External resource disclosure

GraphCast is the only external forecast resource. The 37-level model is documented as trained on ERA5 from 1979 through 2017. The implementation is distributed under Apache-2.0 and the model weights under CC BY-NC-SA 4.0 [10]. Only the required causal forecast fields are sampled; model weights and forecast fields are not redistributed. No third-party operational forecast product is used beyond the official supplied forecasts. Energy-Charts data are used only for the retrospective market context in this report, not for training, inference, siting, or output generation; the API declares a CC BY 4.0 data license [20].

7.3 Verification scope and future improvements

The evaluation combines exact checks with empirical tests. Row coverage, quantile ordering, turbine count, spacing, footprint, and centre depth are verified directly from the generated outputs. Forecast performance, wind-farm yield, and project value are assessed with historical validation, robustness tests, and explicit scenarios. The following points define the scope of those results and identify practical extensions.

Forecast transfer. Chronological validation, worst-year checks, diagnostic-group checks, and the support gate make component selection conservative. They reduce the chance that an average improvement is caused by one favorable year or a narrow subset of rows. The definitive results confirm strong one-day direction performance and identify long-range direction as the clearest opportunity for further improvement. Additional historically causal long-range atmospheric fields would be the most useful extension.

Siting transfer. The farm satisfies all deterministic siting rules. Its expected yield is supported by five annual PyWake replays, nearby-centre tests, bootstrap comparisons, and wind-direction rotations. These complementary checks indicate that the selected shallow-bank design is stable across the available evidence rather than optimized for a single year. Project development would naturally add higher-resolution seabed surveys and further wind observations before construction.

Farm-energy quantiles. The submitted energy estimates are ordered, physically bounded, and generated from the same forecast and layout used elsewhere in the report. They provide a consistent basis for the quantile-bidding calculation. A future extension could jointly model speed and direction uncertainty and validate the resulting farm-power distribution when matched realized farm-power data become available.

Economic scope. The supplied cost model, historical energy replay, and market-price scenarios support transparent comparison of the selected design. Construction schedules, financing structure, taxation, curtailment, and detailed maintenance records are outside the supplied inputs, so they are left as project-specific refinements rather than replaced by arbitrary assumptions. The reported values should therefore be interpreted as comparable planning scenarios.

The most valuable next steps are to strengthen long-range directional information, test the selected shallow-bank neighborhood with additional permitted wind realizations, and update the financial scenarios when project-specific construction and operating data are available.

8 Conclusion

The method connects probabilistic wind prediction, legal wind-farm design, and farm-energy quantiles through a reproducible training and inference repository. Its strongest evidence is specific: complete definitive forecast coverage; paired validation comparisons for all six target branches; a legal 55-turbine design; a same-centre layout gain of 0.49 capacity-factor percentage points with lower wake loss; five-year, directional, bootstrap, and nearby-centre siting tests; and explicit cost, market, bidding, and compute accounting. The code and generation instructions are available at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code`. The complete evaluation supports the selected forecast, site, layout, and bidding policy while also identifying long-range wind direction as the main opportunity for continued improvement.

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